
Richard Nakka's *Experimental Rocketry* Web Site

Rocket Motor Design Charts - Chamber Pressure -

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Introduction

The web page presents *design charts* which may be used to estimate the *steady-state chamber pressure* of a solid rocket motor. A "chamber pressure v.s. Kn chart" is provided for each of the following potassium nitrate based propellants:

- Sorbitol-based KNSB propellant (made with both fine-grind and coarse-grind oxidizer)
- Dextrose-based KNDX propellant
- Sucrose-based KNSU propellant
- Erythritol-based KNER propellant

For the chart data to be considered valid, it is necessary that the propellant be prepared by the "standard" method:

- the propellant must be heat cast
- the oxidizer must be finely ground (except as noted) such that 50% of the particles are less than 75 micron and the remaining 50% less than 250 micron (e.g. fully ground by an electric coffee grinder) Where specified *granular* oxidizer, 90% of the particles are in the range of 300-600 microns.
- the constituents must be very well blended prior to casting (e.g. 1 hour per 100 g. in a rotating mixer)
- be of the standard 65/35 O/F ratio
- the mass density ratio of the cast propellant should be in the range of 94-98% of theoretical

The term *steady-state* infers the operating condition whereby chamber pressure is solely a function of grain burning-surface area. In other words, the generation of combustion gases, and outflow of gases through the nozzle, are in a state of equilibrium

(balance). Therefore, this excludes the initial pressure build-up as well as the pressure tail-off at burnout.

The equations and data that were used to develop the design charts are also presented.

As it is necessary to know Kn of the motor (the ratio of *surface burning area* to *nozzle throat cross-sectional area*) in order to utilize the charts, the methodology for calculation of surface area for a *hollow-cylindrical grain* and *BATES* grain are presented. Examples of such calculations and chart usage are also provided.

Design Charts

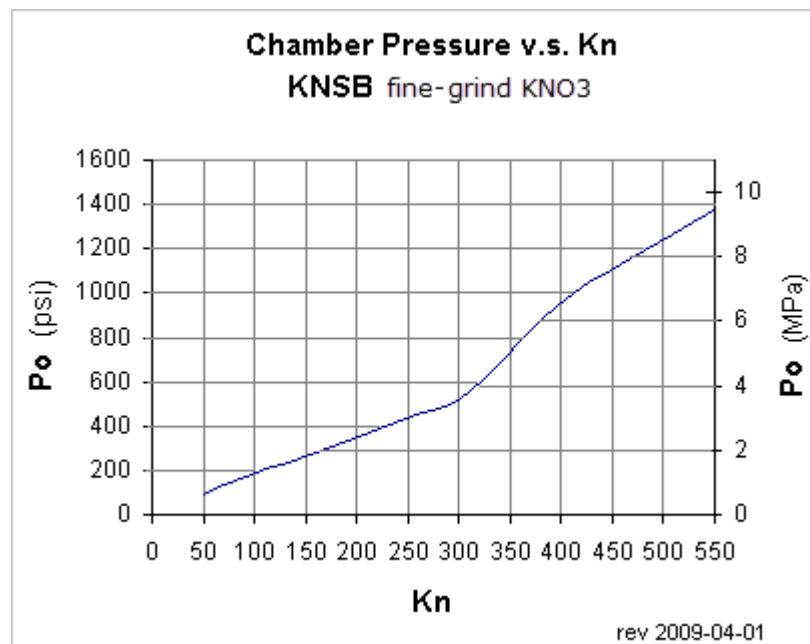


Figure 1-- Design chart for Sorbitol based propellant.

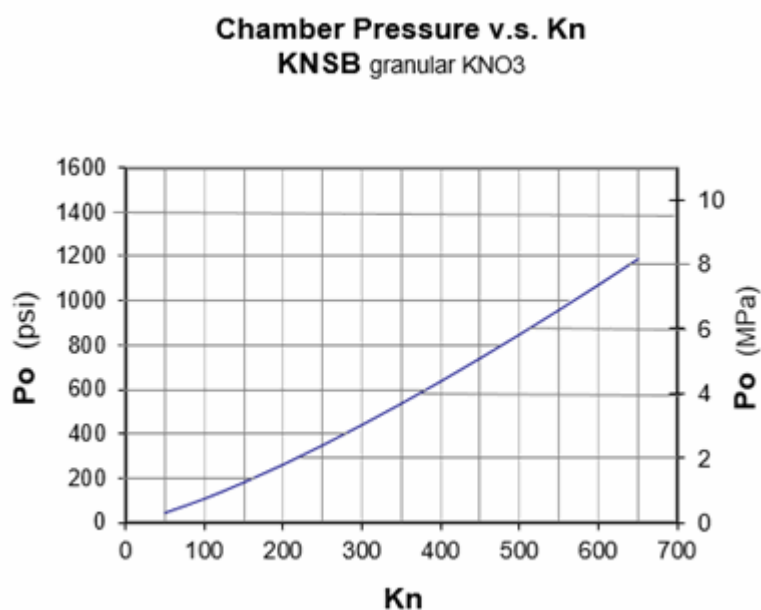


Figure 2-- Design chart for Sorbitol based propellant made with granular oxidizer.

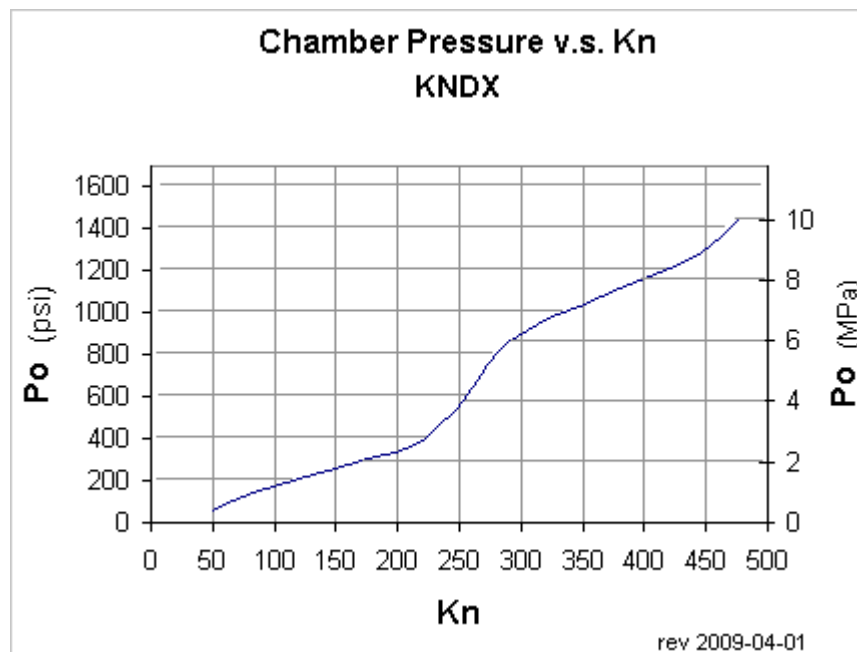


Figure 3-- Design chart for Dextrose based propellant.

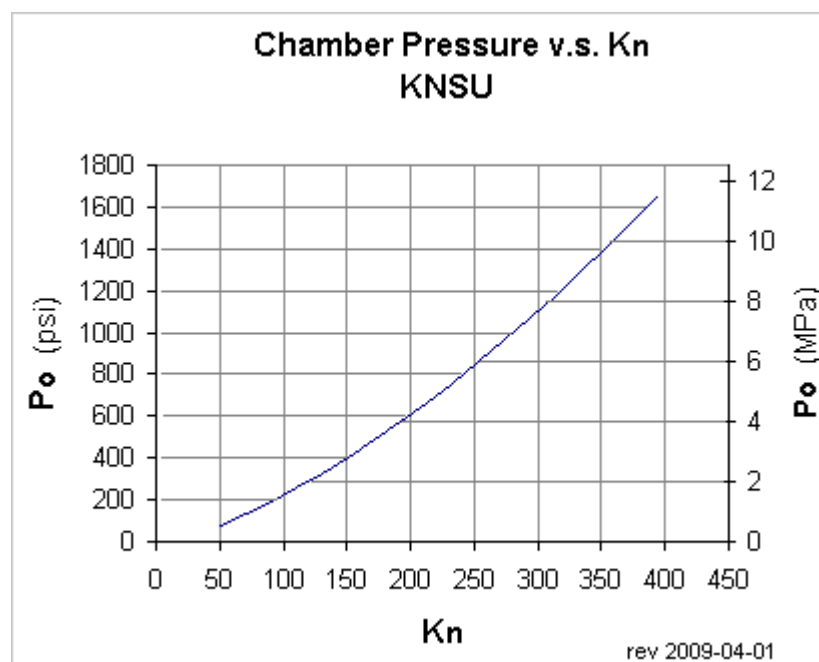


Figure 4-- Design chart for Sucrose based propellant.

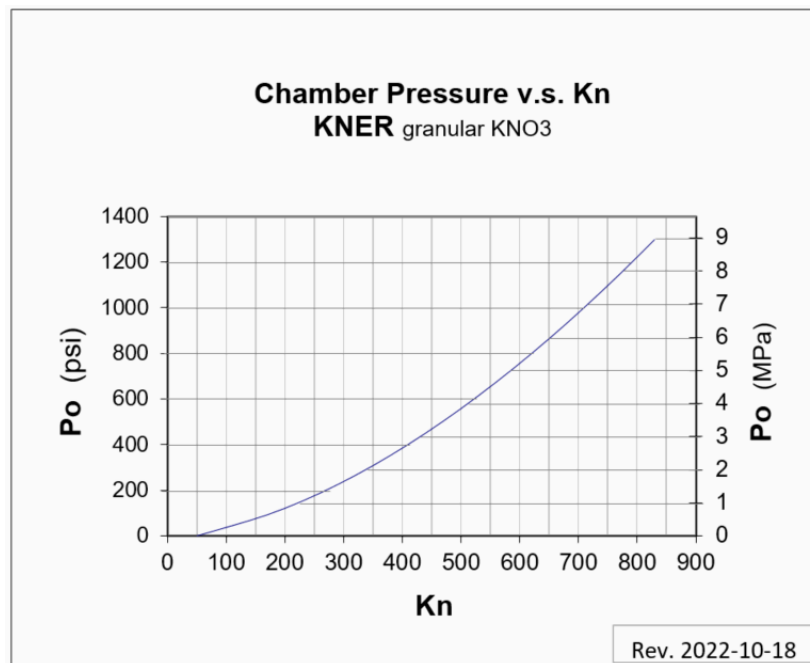


Figure 5-- Design chart for Erythritol based propellant.

Development of Charts

The three design charts were constructed by use of the following expression for steady-state chamber conditions:

$$P_o = \left[K_n \frac{a}{\alpha} \rho c^* \right]^{\frac{1}{1-n}}$$

where the parameters are defined as:

P_o motor chamber pressure

Kn Klemmung, $K_n = A_b / A_t$ (Ratio of Burning surface area (A_b) to the nozzle throat cross-section area (A_t))

a Burn rate pressure coefficient

α Burn rate pressure conversion factor, MPa to Pa units ($\alpha = 1\,000\,000^n$)

ρ propellant mass density

c* propellant characteristic exhaust velocity

n Burn rate pressure exponent

The burn rate pressure coefficients and pressure exponents are based on experimental Strand Burner measurements. Details are provided in the [KN-Dextrose & KN-Sorbitol Propellants -- Burn Rate Experimentation](#) web page. For KNSU, the values used were $a=0.0665$ in/sec (8.26 mm/sec) and $n=0.319$, with these values obtained from an earlier series of Strand Burner measurements. The characteristic exhaust velocity is calculated as shown below. The applicable parameters for each propellant type are given in Tables 1-3:

$$c^* = \sqrt{\frac{R' T_o}{M k} \left(\frac{k+1}{2} \right)^{\frac{k+1}{k-1}}}$$

Rocket Motor chamber pressure as a function of Kn KNSU propellant		
R'	8314 J/mol-K	Universal gas constant
M	41.99 kg/kmol	Effective molecular wt.of products
R	198.0 J/kg-K	Specific gas constant
k	1.133	Ratio specific heats, avg. value
η_c	0.95	Combustion efficiency
To	1720 K	Ideal combustion temperature
To act	1634 K	Actual chamber temperature
c*	895 m/s	Characteristic exhaust velocity
ρ' grain	1.889 g/cm ³	Grain ideal density
ρ/ρ'	0.95	Density ratio (actual/ideal)
ρ grain	1.795 g/cm ³	Grain actual density

Table 1

Rocket Motor chamber pressure as a function of Kn KNDX propellant		
R'	8314 J/kmol-K	Universal gas constant
M	42.39 kg/kmol	Effective molecular wt.of products
R	196.1 J/kg-K	Specific gas constant
k	1.1308	Ratio specific heats, avg. value
η_c	0.95	Combustion efficiency
To	1710 K	Ideal combustion temperature
To act	1625 K	Actual chamber temperature
c*	889 m/s	Characteristic exhaust velocity
ρ' grain	1.879 g/cm ³	Grain ideal density
ρ/ρ'	0.95	Density ratio (actual/ideal)
ρ grain	1.785 g/cm ³	Grain actual density

Table 2

Rocket Motor chamber pressure as a function of Kn KNSU propellant		
R'	8314 J/mol-K	Universal gas constant
M	41.99 kg/kmol	Effective molecular wt.of products
R	198.0 J/kg-K	Specific gas constant
k	1.133	Ratio specific heats, avg. value
η_c	0.95	Combustion efficiency
To	1720 K	Ideal combustion temperature
To act	1634 K	Actual chamber temperature
c*	895 m/s	Characteristic exhaust velocity
ρ' grain	1.889 g/cm ³	Grain ideal density
ρ/ρ'	0.95	Density ratio (actual/ideal)
ρ grain	1.795 g/cm ³	Grain actual density

Table 3

Rocket Motor chamber pressure as a function of Kn KNER propellant		
R'	8314 J/mol-K	Universal gas constant
M	38.78 kg/kmol	Effective molecular wt of products
R	214.4 J/kg-K	Specific gas constant
k	1.139	Ratio specific heats, avg. value
η_c	0.94	Combustion efficiency
T _o	1608 K	Ideal combustion temperature
T _{o act}	1512 K	Actual chamber temperature
c*	894 m/s	Characteristic exhaust velocity
ρ^* grain	1.82 g/cm ³	Grain ideal density
ρ/ρ^*	0.95	Density ratio (actual/ideal)
ρ grain	1.729 g/cm ³	Grain actual density

Table 4

Determination of Kn

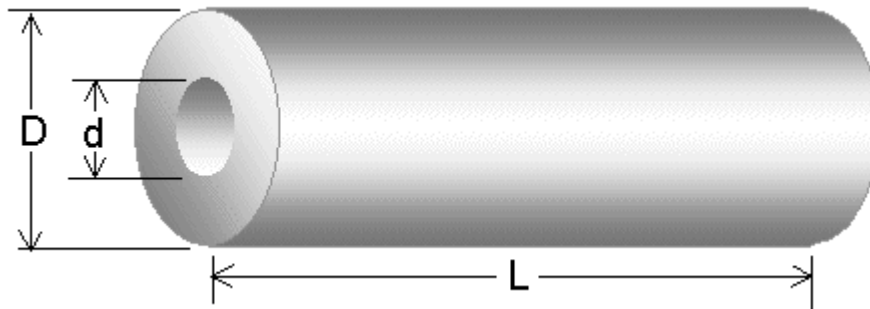


Figure 4-- Hollow cylindrical grain

The burning surface area for a **hollow-cylindrical grain**, as shown in Figure 4, may be calculated as follows:

Grain with unrestricted burning (no surfaces inhibited):

$$A_{b \text{ max}} = A_{b \text{ initial}} = \frac{1}{2} \pi (D^2 - d^2) + \pi L (D + d)$$

$$A_{b \text{ final}} = \pi (D + d) (L - t) \quad \text{where } t = \frac{1}{2} (D - d)$$

Grain with outer surface inhibited (burning at core and ends):

$$A_{b \text{ max}} = A_{b \text{ initial}} = \frac{1}{2} \pi (D^2 - d^2) + \pi d L$$

$$A_{b \text{ final}} = \pi D (L - 2t)$$

Grain with both ends inhibited (burning on outer surface and core):

$$A_b = \text{constant} = \pi L (D + d)$$

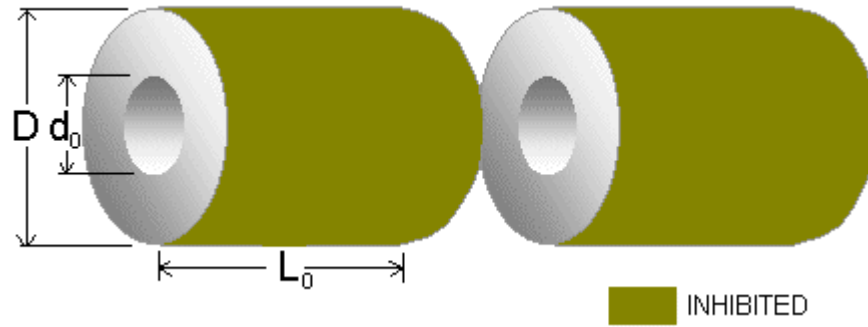


Figure 5-- BATES grain

A **BATES** grain configuration is shown in Figure 5. This usually consists of two or more propellant segments, inhibited on the outer surfaces. This configuration is typically used when a nearly neutral Kn profile is desired (red curve in Figure 6). Kn slope rises to a maximum value then decays. The shape of the curve is determined by the L_0/D and d_0/D ratios. Judicious choice of segment length and core diameter is necessary, or else the Kn profile may instead have only a *progressive* profile (green curve) or *regressive* profile (blue curve). Ideally, Kn_{max} should occur at a surface regression of one-half the web thickness, to produce a symmetric profile (initial Kn = final Kn).

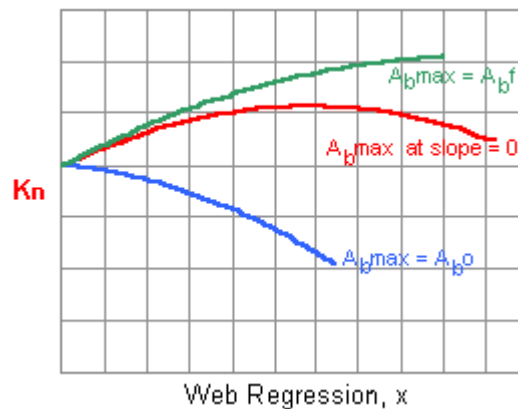


Figure 6-- BATES Kn profiles

The instantaneous grain burning surface area is given by:

$$A_b = N \left[\frac{1}{2} \pi (D^2 - d^2) + \pi L d \right] \quad (\text{Eqn.1})$$

where N is the number of segments; d and L are the instantaneous values of core diameter and segment length, and are given by:

$d = d_0 + 2x$ and $L = L_0 - 2x$ (Eqns.2a, 2b) where x is the linear surface regression (distance the web has burned, normal to the web surface). This is illustrated in Figure 7. The dashed lines represent the geometry of the burning surfaces at some arbitrary point of surface regression.

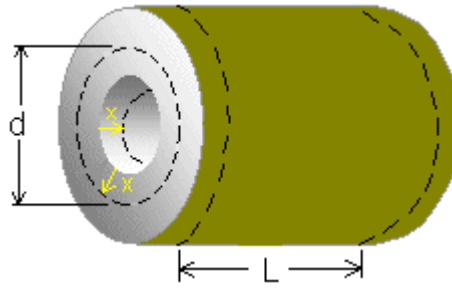


Figure 7-- BATES grain illustrating regression of the web

The initial and final burning surface areas are given by:

$$A_{b \text{ initial}} = N \left[\frac{1}{2} \pi (D^2 - do^2) + \pi Lo do \right]$$

$$A_{b \text{ final}} = N \pi D (Lo - 2t) \quad \text{where } t = \frac{1}{2} (D - do)$$

The value of x when the burning surface area reaches **maximum** is important as this determines maximum chamber pressure. This value of x may be found by setting the derivative (represented by the slope of the Kn v.s. web regression curve) to zero (i.e. $dA_b/dx = 0$), then solving for x .

As such, the value of x is found to be:

$$x = \frac{1}{6} (Lo - 2do) \quad \text{at } A_{b \text{ max}}$$

The value of $A_{b \text{ max}}$ is then found by substituting x into Eqns. 2A and 2B to find d and L , then substituting these values into Eqn. 1.

Note that the Kn profile is *progressive* if the calculation gives $x > do$. In this case, $A_{b \text{ max}} = A_{b \text{ final}}$

The Kn profile is *regressive* if the calculation gives $x < 0$. In this case, $A_{b \text{ max}} = A_{b \text{ initial}}$

When designing a rocket motor, the dimension D is usually limited by factors such as casing or fuselage size. The choice of core diameter, do , is usually based upon desired web thickness (which determines burn time) and erosive burning considerations. Thus, segment length, Lo , is the parameter that may be available to control the Kn profile. The value of Lo may be found which gives a *symmetric Kn profile* (initial and final Kn are equal), if D and do are specified:

$$Lo = \frac{1}{2} (3D + do) \quad \text{for symmetric profile}$$

The "flatness" of the curve, which is always concave downward, is dependant upon the do/D ratio. As do/D approaches unity, the concavity approaches a straight line.

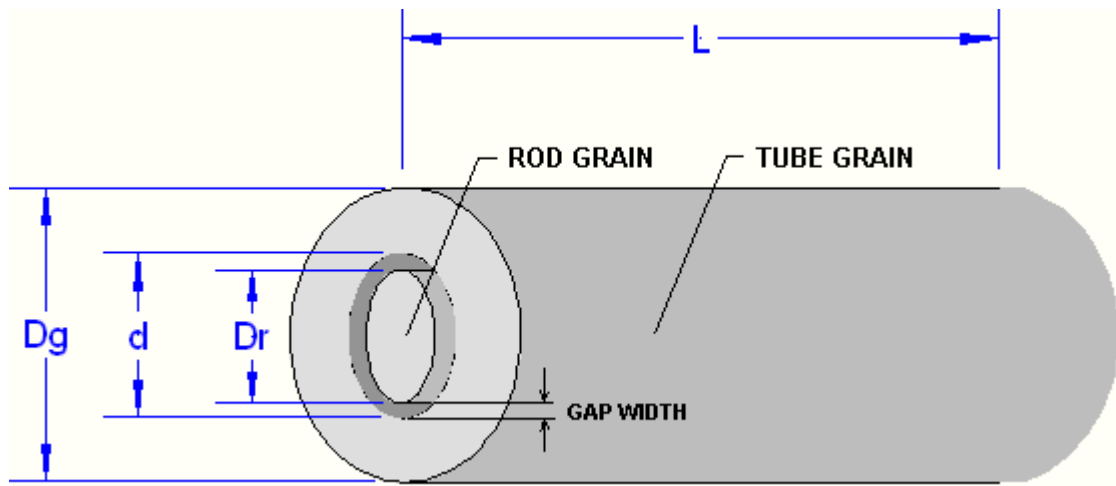


Figure 8-- Rod & Tube grain

The burning surface area for a **Rod & Tube**, as shown in Figure 8, is completely neutral (does not change during the burn) and may be calculated as follows:

$$A_b = \pi (d + D_r) L$$

However, the diameter of the Rod Grain needs to be equal to twice the web thickness of the Tube Grain in order for this to be strictly true. This ensures that burnout of the Rod Grain occurs simultaneously to that of the Tube Grain.

$$D_r = (D_g - d) / 2$$

As well, the following surfaces are inhibited from burning:

- Tube Grain outer surface
- Tube Grain ends (both)
- Rod Grain ends (both)

The gap width should be chosen such that the initial port area (A_p) is a minimum of 2 times the nozzle throat cross-sectional area (A_t).

$$A_p > 2 A_t \quad \text{or}$$

$$\pi / 4 (d^2 - D_r^2) > 2 \pi / 4 D_t^2 \quad (\text{where } D_t \text{ is throat diameter}). \text{ or,}$$

$$(d^2 - D_r^2) > 2 D_t^2$$

Example of Usage

Example 1

Determine the *initial*, *maximum*, and *final* steady-state chamber pressure for an unrestricted hollow cylindrical grain of KNDX with the following

dimensions:

- Outer diameter 2.25 inches
- Core diameter 1.00 inches
- Grain length 10.50 inches

The nozzle throat diameter is 0.650 inches.

Solution:

$$A_{b \max} = A_{b \text{ initial}} = \frac{1}{2} \pi (D^2 - d^2) + \pi L (D + d)$$

This gives $A_{b \max} = A_{b \text{ initial}} = \frac{1}{2} \pi (2.25^2 - 1.00^2) + \pi 10.50 (2.25 + 1.00) = \underline{114 \text{ in}^2}$

$$A_{b \text{ final}} = \frac{1}{2} \pi (D + d) (L - t) \quad \text{where } t = \frac{1}{2} (D - d)$$

Therefore $t = \frac{1}{2} (2.25 - 1.00) = 0.625 \text{ in.}$

and $A_{b \text{ final}} = \pi (2.25 + 1.00) (10.50 - 0.625) = \underline{101 \text{ in}^2}$

The nozzle throat cross-sectional area is: $A_t = \frac{1}{4} \pi (0.650)^2 = 0.332 \text{ in}^2$

This gives an initial, and maximum, $K_n = 114 / 0.332 = \underline{343}$. The final $K_n = 101 / 0.332 = \underline{304}$.

From Figure 2, the initial and maximum steady-state chamber pressure is found to be 1080 psi. The final steady-state chamber pressure is 950 psi.

Example 2

Determine the *initial*, *maximum*, and *final* steady-state chamber pressure for a BATES grain configuration of KNSB with the following dimensions:

- Outer diameter 75 mm.
- Core diameter 22 mm.
- Segment length 100 mm
- 3 segments

The nozzle throat diameter is 13 mm.

Solution:

The value of web regression, x , at the point of maximum chamber pressure is found from the following expression:

$$x = \frac{1}{6} (L_0 - 2d_0)$$

Therefore, $x = \frac{1}{6} [100 - 2(22)] = \underline{9.33 \text{ mm}}$

Substitute the value of x into the following equations:

$$d = d_0 + 2x \quad \text{and} \quad L = L_0 - 2x$$

Giving $d = 22 + 2(9.33) = \underline{40.7 \text{ mm}}$ and $L = 100 - 2(9.33) = \underline{81.3 \text{ mm}}$

Substitute the values of D , L and d into the equation for burning area: $A_b = N$

$$[\frac{1}{2} \pi (D^2 - d^2) + \pi L d]$$

Giving $A_{b \max} = 3 [\frac{1}{2} \pi (75^2 - 40.7^2) + \pi (81.3) 40.7] = \underline{49\,890 \text{ mm}^2}$

The initial and final burning surface areas are given by:

$$A_{b \text{ initial}} = N \left[\frac{1}{2} \pi (D^2 - d_o^2) + \pi L_o d_o \right]$$

$$A_{b \text{ final}} = N \pi D (L_o - 2t) \quad \text{where } t = \frac{1}{2} (D - d_o)$$

This gives:

$$A_{b \text{ initial}} = 3 \left[\frac{1}{2} \pi (75^2 - 22^2) + \pi (100) 22 \right] = \underline{44\,960 \text{ mm}^2}$$

$$\text{Initial web thickness is } t = \frac{1}{2} (75 - 22) = \underline{26.5 \text{ mm}}$$

$$\text{And } A_{b \text{ final}} = 3 \pi 75 [100 - 2(26.5)] = \underline{33\,220 \text{ mm}^2}$$

$$\text{The nozzle throat cross-sectional area is: } A_t = \frac{1}{4} \pi (13)^2 = \underline{133 \text{ mm}^2}$$

The initial, maximum and final Kn can now be calculated:

$$Kn_{\text{initial}} = 44\,960 / 133 = \underline{338}$$

$$Kn_{\text{max}} = 49\,890 / 133 = \underline{375}$$

$$Kn_{\text{final}} = 33\,220 / 133 = \underline{250}$$

From Figure 2, the maximum steady-state chamber pressure is found to be 6.3 MPa. The initial and final steady-state chamber pressure is 5.0 MPa and 3.1 MPa, respectively.

Example 3

A rocket motor with a BATES grain configuration of KNDX is to be designed to give an approximately neutral burn profile, with matching initial and final Kn values (producing the maximum Kn midway through the web regression). For the following segment and core diameters, what length should the segments initially be?

- Outer diameter 50 mm.
- Core diameter 18 mm.
- 4 segments

Solution:

$$L_o = \frac{1}{2} (3D + d_o)$$

$$\text{This gives } L_o = \frac{1}{2} [3(50) + 18] = \underline{84 \text{ mm}}.$$

Verify that initial and final surface areas are identical:

$$A_{b \text{ initial}} = 4 \left[\frac{1}{2} \pi (50^2 - 18^2) + \pi (84) 18 \right] = \underline{32\,673 \text{ mm}^2}$$

$$A_{b \text{ final}} = 4 \pi 50 [84 - 2(16)] = \underline{32\,673 \text{ mm}^2} \quad \text{where } t = \frac{1}{2} (50 - 18) = 16 \text{ mm}$$



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